



Smart Path: Robot for Line Following, Color Detection & Obstacle Avoidance

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Abstract

The Codey Rocky robot and the mBlock programming platform with Python language are used in this project to construct an autonomous mobile robot. The three primary parts of the robot are color identification, obstacle avoidance and line tracking. The project focuses on developing control algorithms based on sensor inputs rather than building the robot hardware. All the features have been tested and developed effectively. Using buttons that each engage a distinct function, the system enables switching between various modes. The robot is able to follow a line, detect obstacles and recognize different colors and display them. The outcomes show that the system functions independently with acceptable accuracy and dependability.

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Introduction

Modern engineering applications like industrial automation and intelligent systems make expensive use of autonomous robots. In order to observe their surroundings and make decisions without human assistance, these systems rely on sensors and control algorithms. Codey Rocky, a pre-built robot is programmed in Python mode using the mBlock platform to provide autonomous behavior. Creating a system that allows the robot to recognize colors, follow a specific line and avoid obstacles is the primary goal. This project demonstrates the application of the theoretical concepts that applied in mechatronics and robotics in a practical and real-world scenario.

Background

Line-following robots use sensors to distinguish between a white surface and a black line. The robot adjusts its movement to stay on track based on sensor data. Distance sensors, which help the robot detect objects and change course to avoid collisions, are used in obstacle avoidance. The robot can identify various colors and respond appropriately by detecting them, thanks to color detection. An essential concept in autonomous robotics, these systems depend on control algorithms that analyze sensor inputs and make decisions in real time.

Methodology

The Codey Rocky robot was designed to carry out autonomous activities using the mBlock programming environment in Python mode. The robot has sensors that supply the input data needed to regulate its actions. There were three primary phases to the system's development. Initially, a line-following algorithm was used, in which the robot modifies its course according to the path it detects. Second, distance sensing was used to implement obstacle avoidance, which enables the robot to identify objects and alter its course to prevent collisions. Third, the robot was able to identify and show colours thanks to the development of colour detection. Additionally, buttons were used to designate various control modes: button A initiates the colour detecting mode, button B initiates the obstacle avoidance mode, and button C initiates the line-following mode. This method makes it simple to swap between features.

Flowchart

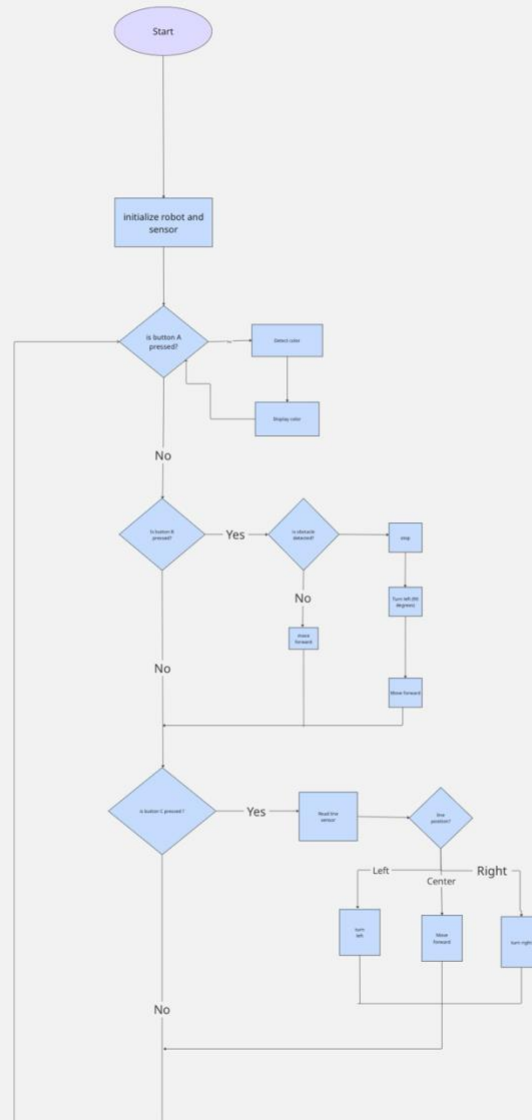


Figure 1: Flowchart of the robot control system using button-based modes(A ; color detection, B: obstacle avoidance, and C: line following).

System Implementation

The system uses Python to combine all three functions into a single control program. The robot acts according to the chosen mode while continuously reading sensor data. The robot enters line-following mode and modifies its movement to remain on the path when button C is pressed. The robot enters obstacle avoidance mode when button B is pressed, detecting objects and altering its course to prevent collisions. The robot enters color detecting mode and shows the identified colour on its screen when button A is pressed. This modular architecture guarantees efficiency and adaptability.

Results and Discussion

The team has programmed Codey Rocky using mBlock 5, which is a fusion of block-based and Python. The three core functionalities have been developed and tested individually. Color detection that identifies red, blue, yellow, green, cyan, black and white through a built-in color sensor. Line tracking algorithm using the infrared sensors to follow a black line on a white surface; and an obstacle avoidance routine using the sensor built-in to stop, turn, and reverse when needed as soon as an obstacle is detected. These functionalities have been verified to work reliably under controlled conditions. Initial tests demonstrate that each function performs well in isolation. Color detection works accurately most of the time unless the light setting is affected. The solution is to color calibrate using mBlock 5 for the color sensor to be adjusted. Line tracking follows a path with average speed while obstacle avoidance is triggered at 5cm reliably. However, when integrated together. The robot may avoid the obstacle but takes approximately 30 seconds relocating the line.

Project Management

The workload has been divided equally among the two members. One member leads the integration of simultaneous functions and line tracking optimization while the other is responsible for color detection calibration and developing obstacle avoidance logic. All members collaborate on testing, documentation, and troubleshooting integration conflicts during biweekly meetings.

Challenges and Solutions

Three main challenges were encountered. Firstly, the programming aspect and the logic were complicated regarding integrating multiple functions into one. Secondly, resource limitations, the microprocessor has limited processing speed, causing a lag especially when two functions are integrated. Finally, tasks conflict. For instance, obstacle avoidance overrides the path following which causes the robot to drift off course. Those challenges encourage the members to develop their problem-solving skills and to divide and conquer.

Future Work

All project objects have been successfully achieved. The robot is capable of performing color detection, line tracking, and obstacle avoidance. The team members will test the robot on different arenas to optimize the code and adjust the motor speed in order to decrease overshoot after obstacle avoidance maneuvers. The team shall document and provide the final comments and flowchart while recording a demo video of the robot performing all the core functionalities successfully.

Self Reflection

Working on this project taught us many lessons beyond coding. First, we learned that integrating multiple functions is significantly harder than making each work individually. The integrations of several functionalities offered a larger challenge that allowed us to solve on a larger scale rather than isolated systems. Skills evolved from simple coding into integration of priority-based systems. Teamwork strengthened through meetings and mutual problem-solving. Patience and persistence grew after many trial-and-error attempts, transforming us into real problem-solvers capable of delivering a working embedded robotic system.

If we could start this project over, we would do three things differently. First, the online learning environment slowed down our progress. If we had known ahead of time that the job would be completed primarily online, we would have begun work sooner and better controlled our time. This would have helped us finish faster and with fewer delays.

Second, we would have learned earlier how to use Codey Rocky with the iPad and connect it via Bluetooth. Using the iPad made programming easier and quicker than other methods. If we had discovered this sooner, we could have saved time and made our work more efficient.

Finally, we would have improved our skills in Python programming before starting the project. We faced some challenges when trying to combine obstacle avoidance and line following. Having a stronger background in Python would have helped us solve these problems more easily and build a better system.

Conclusion

The Codey Rocky Project successfully achieved color detection and simultaneous line tracking and obstacle avoidance. Challenges overcome included sensor interference, multiple functions integration and requirements adjustment. Limitations remain with sharp turns and several light settings. The project met all objectives, and the team gained valuable hands-on experience in real-time coordination and skills that are applicable in future such as systematic debugging.

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